



Problem

Given a value V , you have an infinite supply of N different coin denominations. Find the minimum number of coins required to make exactly V cents.

You must also print the coins used in an optimal solution.

If it is not possible to make the value V , print -1.

Input Format

The first line contains an integer T , the number of test cases.

For each test case: The first line contains two integers V and N .

V — the target value. N — the number of available coin denominations. The second line contains N integers $C[i]$, representing the coin denominations.

Constraints

$1 \leq T \leq 100$ $1 \leq V \leq 10^6$ $1 \leq N \leq 10^6$ $1 \leq C[i] \leq 10^6$

Line: 16 Col: 17

Output Format

For each test case:

Print the coins used in a minimum-coin solution, separated by spaces. In the next line, print the minimum number of coins. If it is impossible to make V , print -1.

Sample Input 0

```
7 2
2 4
```

Sample Output 0

```
-1
```

Sample Input 1

```
11 4
1 5 6 9
```

Sample Output 1

```
2
```

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Language

Python 3



```
1 # Enter your code here. Read input from STDIN.
2 v,n=map(int,input().split())
3 coins=list(map(int,input().split()))
4
5 dp=[999999]*(v+1)
6 dp[0]=0
7
8 for i in range(1,v+1):
9     for c in coins:
10         if c<=i:
11             dp[i]=min(dp[i],dp[i-c]+1)
12
13 if dp[v]==999999:
14     print(-1)
15 else:
16     print(dp[v])
```



Upload Code as File

Run Code

Submit Code



Test against custom input

Congratulations

You solved this challenge. Would you like to challenge your friends?



Test case 0



Test case 1

Compiler Message

Success